

### 1. Coronary Artery Disease (CAD)

**Pathophysiology:** CAD is characterized by the chronic buildup of atherosclerotic plaques within the coronary arteries. This process begins with endothelial injury followed by the accumulation of low-density lipoproteins (LDL) and inflammatory cells in the vessel wall. **Etiology:** Risk factors include smoking, diabetes, hypertension, and dyslipidemia (the "SAD CHF" mnemonic). **Mechanism:** Plaque progression causes luminal stenosis, reducing myocardial oxygen supply. Complete occlusion via plaque rupture and subsequent thrombosis results in myocardial infarction (heart attack). **Diagnostic Indicators:** Common features for ML models include age, sex, chest pain (angina), EKG changes, and biomarkers like troponin. **Management:** Treatment involves lifestyle modification, antiplatelet therapy (aspirin), statins, and invasive procedures like angioplasty or coronary artery bypass grafting.

### 2. Stroke (Cerebrovascular Accident)

**Pathophysiology:** A stroke occurs when blood supply to the brain is disrupted, leading to rapid neuronal death. There are two main types: Ischemic (87% of cases), caused by a clot, and Hemorrhagic, caused by a ruptured vessel. **Clinical Symptoms:** Presentation often follows the "BE FAST" acronym—Balance loss, Eyes (vision changes), Face drooping, Arm weakness, Speech difficulty, and Time to act. **Mechanisms:** Ischemic strokes involve thrombotic or embolic occlusions; hemorrhagic strokes involve intracerebral or subarachnoid bleeding, often due to hypertension. **Diagnosis:** Primarily confirmed via CT or MRI imaging to distinguish between clot and bleed. **Treatment:** Ischemic stroke requires rapid thrombolysis (tPA) within 4.5 hours or mechanical thrombectomy. Hemorrhagic stroke focus is on blood pressure control and intracranial pressure reduction.

### 3. Diabetes Mellitus (Type 1 and 2)

**Pathophysiology:** A chronic metabolic disorder defined by hyperglycemia due to defects in insulin secretion, insulin action, or both. **Type 1 (T1DM):** An autoimmune condition where the immune system destroys insulin-producing beta cells in the pancreas, leading to absolute insulin deficiency. **Type 2 (T2DM):** Primarily characterized by insulin resistance and relative insulin deficiency, strongly linked to obesity and lifestyle. **Symptoms:** Common "3 Ps"—Polyuria (frequent urination), Polydipsia (excessive thirst), and Polyphagia (extreme hunger), alongside blurred vision and fatigue. **Complications:** Long-term damage includes retinopathy, nephropathy, neuropathy, and increased cardiovascular risk. **Management:** T1DM requires lifelong insulin; T2DM is managed with lifestyle changes, oral agents like Metformin, and sometimes insulin.

### 4. Chronic Obstructive Pulmonary Disease (COPD)

**Pathophysiology:** A progressive, irreversible airflow limitation caused by an abnormal inflammatory response to noxious particles, typically cigarette smoke. **Components:** Includes Emphysema (destruction of air sacs/alveoli) and Chronic Bronchitis (inflammation of the bronchial tubes with mucus hypersecretion). **Mechanism:** Loss of elastic recoil and small airway fibrosis lead to "air trapping" and lung hyperinflation. **Clinical Signs:** Chronic cough, sputum production, and exertional dyspnea (shortness of breath). **Diagnosis:** Post-bronchodilator spirometry showing an FEV1/FVC ratio  $< 0.70$  is the gold standard. **Management:** Smoking cessation is critical; pharmacological treatment includes bronchodilators, inhaled corticosteroids, and oxygen therapy in advanced stages.

### 5. Alzheimer's Disease

**Pathophysiology:** A neurodegenerative disorder marked by the accumulation of extracellular amyloid-beta plaques and intracellular tau tangles in the brain. **Mechanism:** These protein abnormalities disrupt synaptic communication and lead to progressive neuronal loss, beginning in the hippocampus (critical for memory). **Etiology:** While most cases are sporadic and age-related, genetic factors like the APOE- $\epsilon 4$  allele increase risk. **Clinical Presentation:** Early signs include short-term memory loss, followed by disorientation, mood swings, and loss of language skills. **Stages:** Progresses from mild cognitive impairment to severe dementia, eventually impacting basic bodily functions like swallowing. **Diagnosis:** Based on clinical history, cognitive testing, and exclusion of other causes via brain imaging (MRI/PET). **Treatment:** No cure exists; medications like cholinesterase inhibitors (e.g., Donepezil) provide modest symptomatic relief.

### 6. Lung Cancer

**Pathophysiology:** Uncontrolled cellular proliferation within lung tissues, primarily categorized into Small Cell (SCLC) and Non-Small Cell Lung Cancer (NSCLC). **Etiology:** Tobacco smoke

is the leading cause (85-90% of cases), though radon exposure and air pollution are significant factors. Mechanism: Carcinogens cause DNA mutations in bronchial epithelial cells, leading to tumor formation and potential metastasis via the lymphatic system. Symptoms: Persistent cough, hemoptysis (coughing up blood), chest pain, and unexplained weight loss. Diagnosis: Identified through chest X-rays or CT scans and confirmed by tissue biopsy. Staging: Uses the TNM system (Tumor, Node, Metastasis) to determine severity. Management: Treatment plans often involve surgical resection, chemotherapy, radiation, or targeted immunotherapy depending on the cancer subtype.

#### 7. Influenza (The Flu)

Pathophysiology: An acute respiratory infection caused by Influenza A and B viruses. Transmission: Highly contagious, spreading via respiratory droplets from coughing or sneezing. Mechanism: The virus targets and destroys the respiratory epithelial lining, triggering a systemic inflammatory response. Symptoms: Sudden onset of high fever, myalgia (muscle aches), headache, severe malaise, and non-productive cough. Complications: Can lead to primary viral pneumonia or secondary bacterial infections like *Streptococcus pneumoniae*. Diagnosis: Rapid molecular assays (RT-PCR) are preferred for definitive detection. Prevention/Management: Annual vaccination is the primary prevention strategy; antiviral medications (e.g., Oseltamivir) are effective if administered within 48 hours of symptom onset.

#### 8. Tuberculosis (TB)

Pathophysiology: A bacterial infection caused by *Mycobacterium tuberculosis*. Mechanism: The bacteria are inhaled into the lungs, where they are engulfed by alveolar macrophages. If the immune system fails to contain them, "latent TB" progresses to "active TB," forming granulomas. Clinical Presentation: Characterized by a persistent cough (often for >3 weeks), night sweats, fever, and weight loss. Extrapulmonary TB: The infection can spread to kidneys, spine (Pott's disease), or the central nervous system. Diagnosis: Confirmed via Sputum Acid-Fast Bacilli (AFB) smear, culture, or Chest X-ray showing apical infiltrates. Management: Requires a multi-drug regimen (e.g., RIPE: Rifampin, Isoniazid, Pyrazinamide, Ethambutol) for at least 6 months to prevent drug resistance.

#### 9. Malaria

Pathophysiology: A febrile illness caused by *Plasmodium* parasites transmitted via the bite of an infected female *Anopheles* mosquito. Mechanism: Parasites travel to the liver to mature and then infect red blood cells (RBCs). The rupture of these RBCs triggers cyclical fevers. Species: *P. falciparum* is the most dangerous, potentially causing cerebral malaria. Symptoms: Characterized by "paroxysms" of chills followed by high fever and profuse sweating. Anemia and jaundice are common due to RBC destruction. Diagnosis: Gold standard is microscopic examination of thick and thin blood smears. Management: Artemisinin-based combination therapies (ACTs) are the standard treatment; prevention includes insecticide-treated nets and chemoprophylaxis for travelers.

#### 10. COVID-19

Pathophysiology: Caused by the SARS-CoV-2 virus, which enters cells via the ACE2 receptor. Mechanism: Primarily affects the lower respiratory tract, potentially triggering a "cytokine storm"—a severe hyper-inflammatory state. Clinical Manifestations: Range from asymptomatic to severe pneumonia and Acute Respiratory Distress Syndrome (ARDS). Common symptoms include cough, fever, loss of taste/smell (anosmia), and fatigue. Long-COVID: Some patients experience lingering neurological or cardiovascular symptoms for months post-infection. Diagnosis: Confirmed via nucleic acid amplification tests (RT-PCR) or rapid antigen tests. Management: Treatment includes supportive care, corticosteroids for severe cases, and antivirals; global control is largely driven by large-scale mRNA and viral-vector vaccination programs.